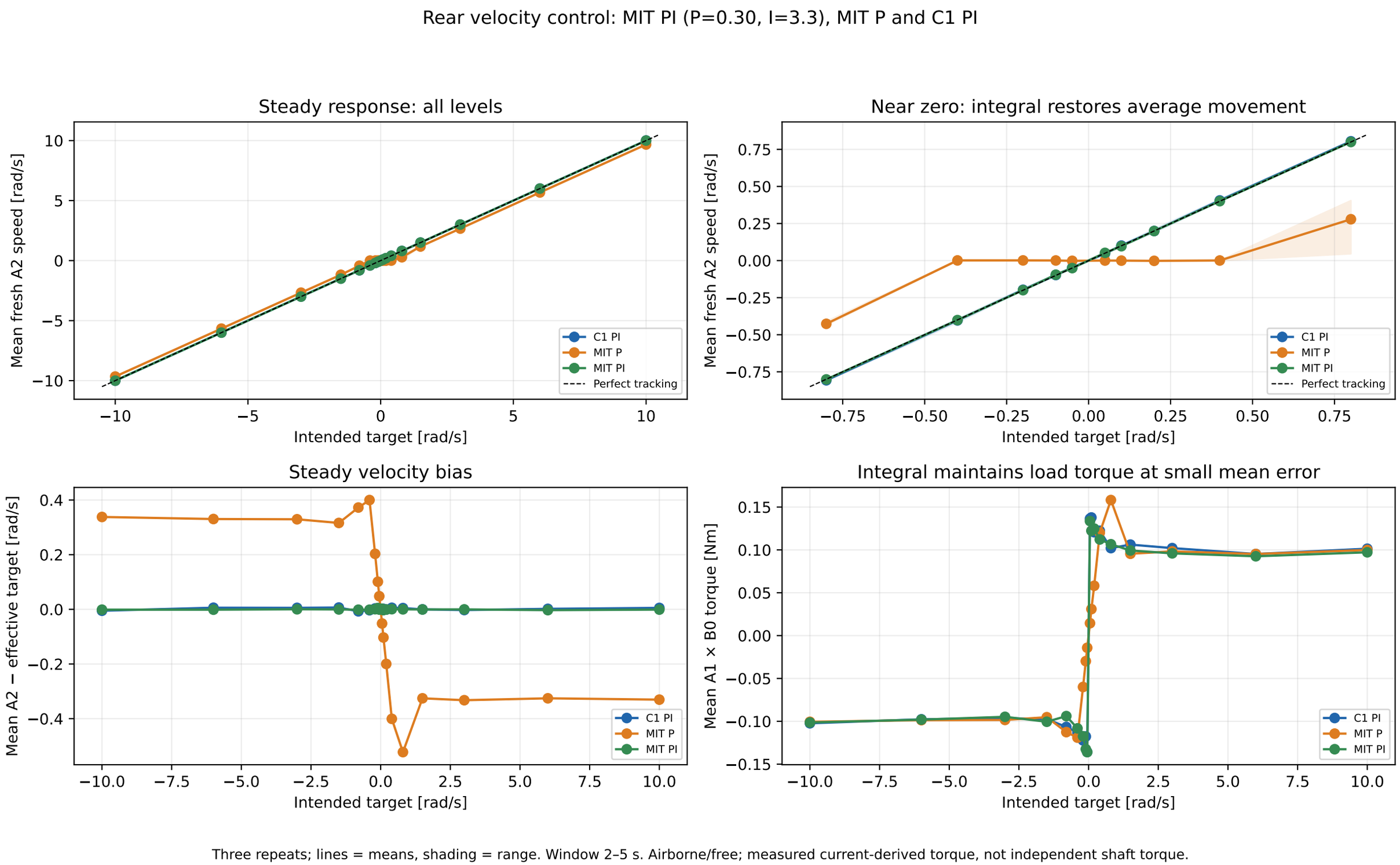
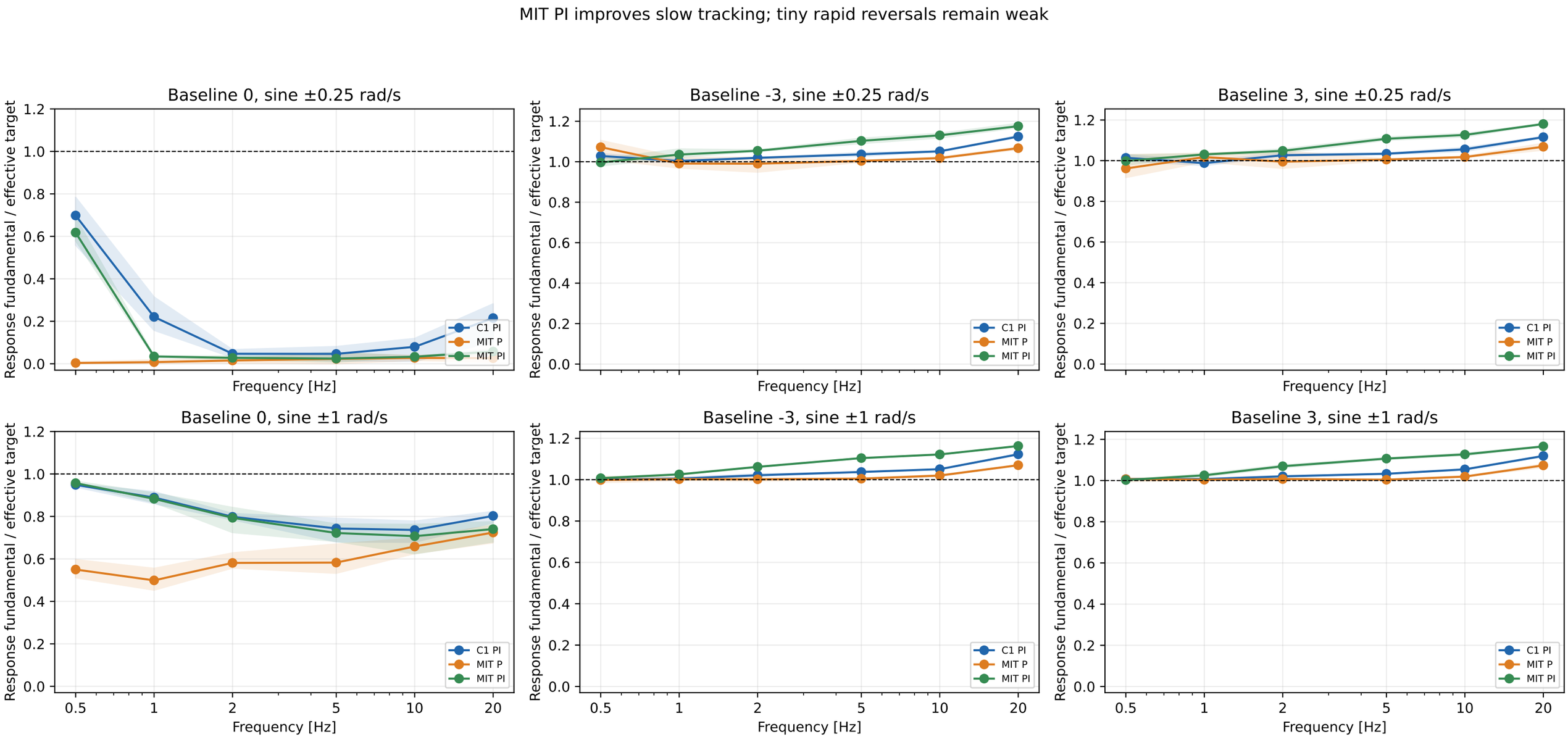
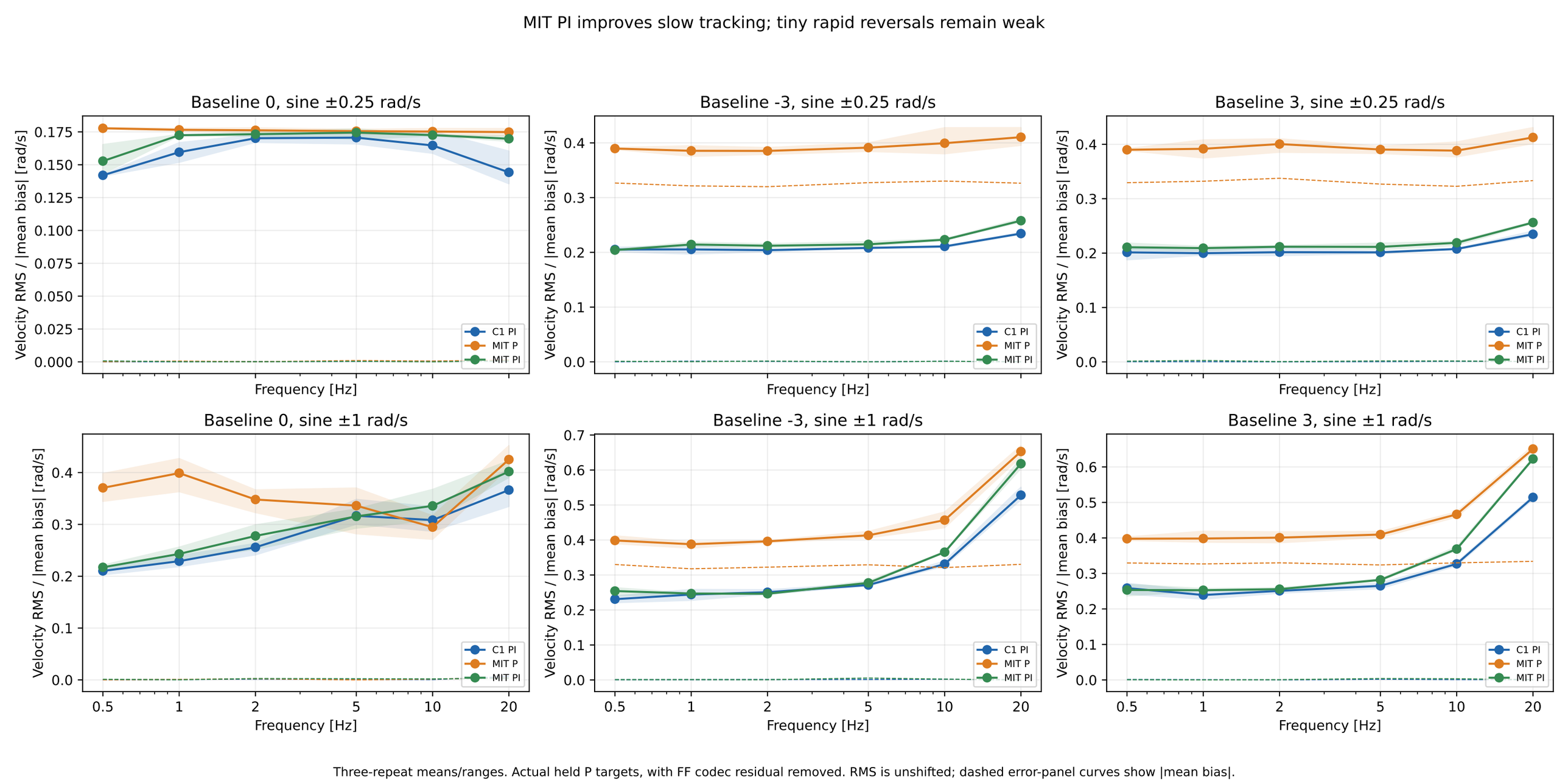
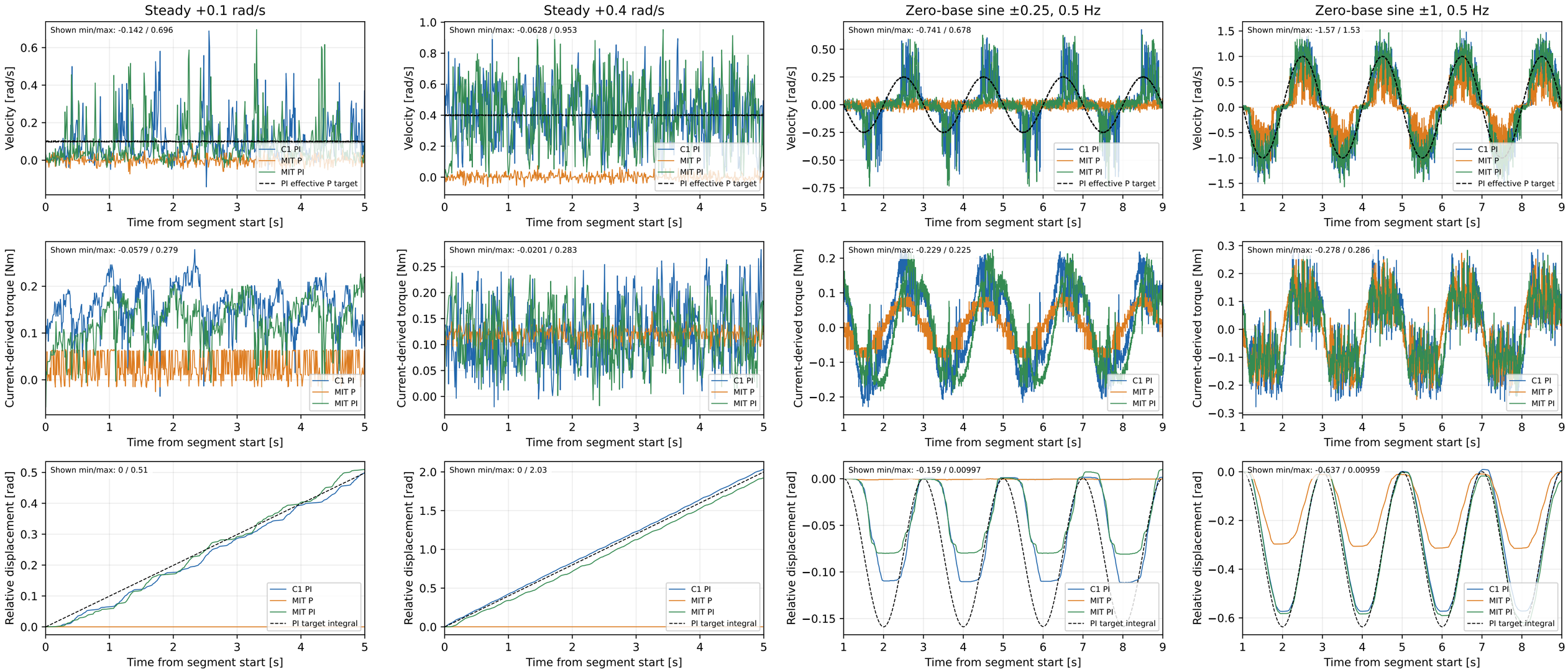
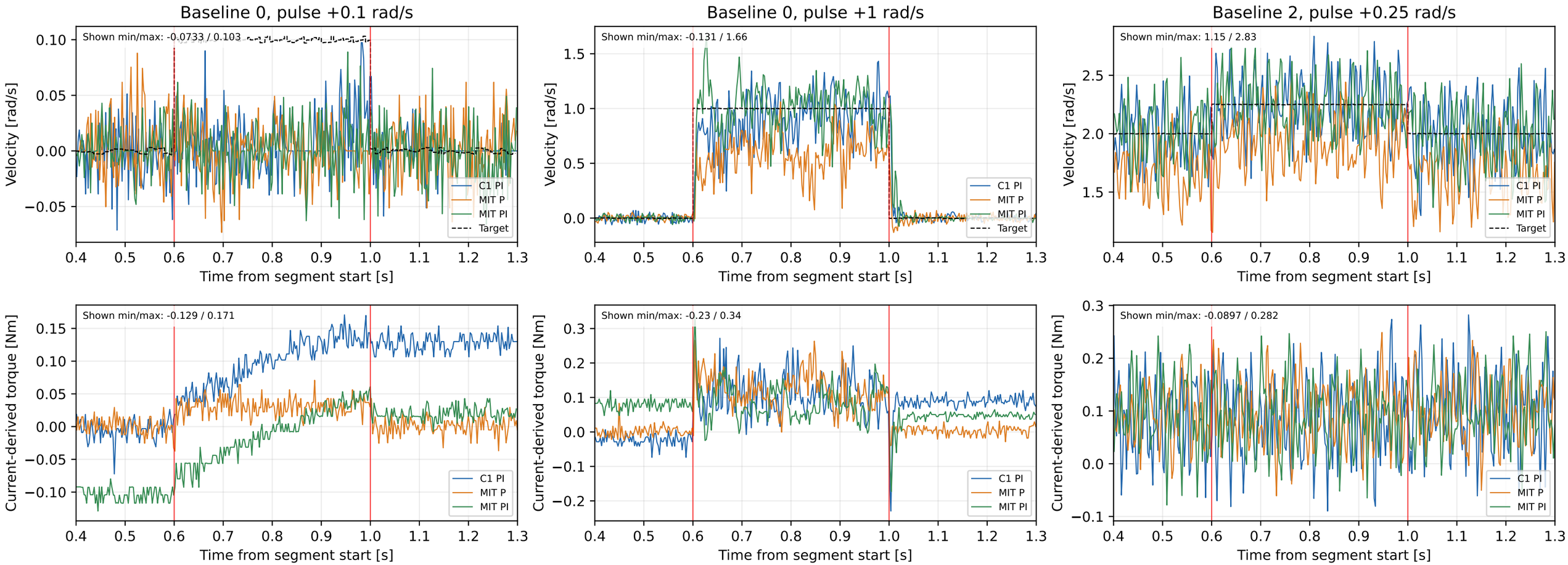
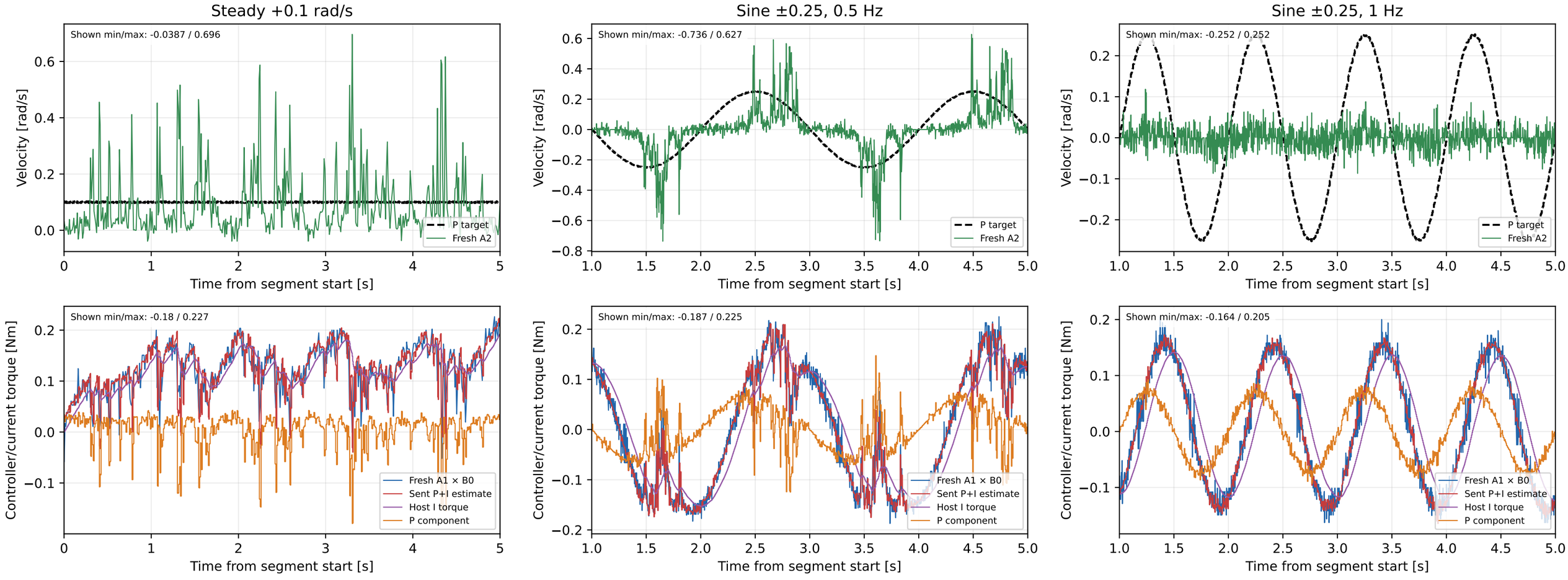

Rear velocity control: MIT PI (P=0.30, I=3.3), MIT P and C1 PI
Steady response: all levels
Mean fresh A2 speed [rad/s]
Intended target [rad/s]
C1 PI
MIT P
MIT PI
Perfect tracking
Near zero: integral restores average movement
Mean fresh A2 speed [rad/s]
Intended target [rad/s]
C1 PI
MIT P
MIT PI
Perfect tracking
Steady velocity bias
Mean A2 − effective target [rad/s]
Intended target [rad/s]
C1 PI
MIT P
MIT PI
Integral maintains load torque at small mean error
Mean A1 × B0 torque [Nm]
Intended target [rad/s]
C1 PI
MIT P
MIT PI
Three repeats; lines = means, shading = range. Window 2–5 s. Airborne/free; measured current-derived torque, not independent shaft torque.


MIT PI improves slow tracking; tiny rapid reversals remain weak
Baseline 0, sine ±0.25 rad/s
Baseline -3, sine ±0.25 rad/s
Baseline 3, sine ±0.25 rad/s
Baseline 0, sine ±1 rad/s
Baseline -3, sine ±1 rad/s
Baseline 3, sine ±1 rad/s
Response fundamental / effective target
Frequency [Hz]
C1 PI
MIT P
MIT PI
Three-repeat means/ranges. Actual held P targets, with FF codec residual removed. RMS is unshifted; dashed error-panel curves show |mean bias|.


MIT PI improves slow tracking; tiny rapid reversals remain weak
Baseline 0, sine ±0.25 rad/s
Baseline -3, sine ±0.25 rad/s
Baseline 3, sine ±0.25 rad/s
Baseline 0, sine ±1 rad/s
Baseline -3, sine ±1 rad/s
Baseline 3, sine ±1 rad/s
Velocity RMS / |mean bias| [rad/s]
Frequency [Hz]
C1 PI
MIT P
MIT PI
Three-repeat means/ranges. Actual held P targets, with FF codec residual removed. RMS is unshifted; dashed error-panel curves show |mean bias|.


Matched targets, repeat 3: native speed, current-derived torque and encoder travel
Steady +0.1 rad/s
Steady +0.4 rad/s
Zero-base sine ±0.25, 0.5 Hz
Zero-base sine ±1, 0.5 Hz
Shown min/max: -0.142 / 0.696
Shown min/max: -0.0628 / 0.953
Shown min/max: -0.741 / 0.678
Shown min/max: -1.57 / 1.53
Velocity [rad/s]
Shown min/max: -0.0579 / 0.279
Shown min/max: -0.0201 / 0.283
Shown min/max: -0.229 / 0.225
Shown min/max: -0.278 / 0.286
Current-derived torque [Nm]
Shown min/max: 0 / 0.51
Shown min/max: 0 / 2.03
Shown min/max: -0.159 / 0.00997
Shown min/max: -0.637 / 0.00959
Relative displacement [rad]
Time from segment start [s]
C1 PI
MIT P
MIT PI
PI effective P target
PI target integral
Separate runs; no smoothing or fitted time/bias shift. Each A3 subtracts its own first shown sample. Same intended targets; the plotted reference is PI's codec-corrected P target.


Matched step responses, repeat 3; red lines bound the 0.4-s pulse
Baseline 0, pulse +0.1 rad/s
Shown min/max: -0.0733 / 0.103
Baseline 0, pulse +1 rad/s
Shown min/max: -0.131 / 1.66
Baseline 2, pulse +0.25 rad/s
Shown min/max: 1.15 / 2.83
Velocity [rad/s]
Current-derived torque [Nm]
Shown min/max: -0.129 / 0.171
Shown min/max: -0.23 / 0.34
Shown min/max: -0.0897 / 0.282
Time from segment start [s]
C1 PI
MIT P
MIT PI
Target
Shown 0.4–1.3 s to resolve the pulse after the initial baseline transient. Integral resets once per segment. Native A2/A1; no smoothing/time shift.


MIT PI torque decomposition: integral supports movement but cannot cure every reversal
Steady +0.1 rad/s
Sine ±0.25, 0.5 Hz
Sine ±0.25, 1 Hz
Shown min/max: -0.0387 / 0.696
Shown min/max: -0.736 / 0.627
Shown min/max: -0.252 / 0.252
Velocity [rad/s]
Time from segment start [s]
P target
Fresh A2
Shown min/max: -0.18 / 0.227
Shown min/max: -0.187 / 0.225
Shown min/max: -0.164 / 0.205
Controller/current torque [Nm]
Fresh A1 × B0
Sent P+I estimate
Host I torque
P component
Command estimates use latest A2 at send; A1 is independently queried, unsmoothed. Known-law held-out current R²=0.917; this is not forward plant prediction.